

## Document 2: The Decision Log (Module 1)

- **Project Kickoff:** Established an environment baseline for a Windows 11 machine with 8GB RAM. Confirmed that local hardware limitations are bypassed because the AI computation runs entirely on Google's cloud servers via API.
- **Environment Isolation:** Encountered a `KeyboardInterrupt` during the initial setup of the virtual environment (`venv`). Solved it by deleting the corrupted folder and re-running `python -m venv venv` patiently, allowing the background `pip` installation to conclude cleanly.
- **Workspace Synchronization:** Encountered a VS Code alert indicating that terminal environment injection was disabled. Resolved this by adjusting the global configuration setting `python.terminal.useEnvFile` to `True`, ensuring VS Code and the command line perfectly sync environment files.
- **Handling Server Spikes:** Hit an initial network error `503 UNAVAILABLE` due to global traffic spikes on Google's endpoint. Verified that our script structure was correct, implemented a basic manual retry tactic, and successfully established a stable connection on the subsequent attempt.
- **Structural Transformation:** Shifted the AI model from a standard text generator to a deterministic data output engine by introducing a target `response_mime_type="application/json"` configuration block.